Memory Companion

An AI companion for people living with dementia — and the families who care for them

Nebius x NVIDIA Global AI Hackathon 2026 | Track: Personal AI | Submit by Oct 29 (deadline: Oct 30, 10:00am PDT / 10:30pm IST)

In one line: A tablet companion that helps a person with dementia recognize their loved ones, answers their repeated questions patiently, calms them when they are distressed, and keeps the family informed — running on open NVIDIA models that the family controls.

1. The problem

Tens of millions of people worldwide live with dementia. They forget faces, repeat the same questions many times a day, become confused and anxious in the evenings, and sometimes wander and get lost. They are also prime targets for scammers. Family carers burn out answering the same questions again and again, worry constantly, and have little information to share with doctors.

Most existing tools are either simple reminder apps or general-purpose assistants. None are built around the person's own family, memories, and the emotionally difficult moments that define daily life with dementia.

2. Hackathon requirements

- Project must run on **Nebius Token Factory** or **Nebius AI Cloud**.
- Must use **at least one NVIDIA open-source model** (Nemotron is our core model).
- Submission needs a working project, demo video, and **feedback** on the Nebius and NVIDIA tools used.
- Entered in the **Personal AI** track — a private AI the family owns and controls.

3. Features

Core features (must ship)

#	Feature	What it does
1	"Who is this?"	Patient points the camera at someone; companion answers warmly with context: "That's Rohan, your grandson. He loves cricket — ask him about his match." Never guesses when unsure.
2	Patient answers to repeated questions	Answers "What day is it?" or "When is lunch?" the tenth time as kindly as the first, in the patient's language, using family-provided information.
3	Family-written answers to painful questions	Family decides in advance how to respond when the patient asks for someone who has died: gently redirect, share a happy memory, or call a relative. Our most original feature.
4	Daily family digest	Evening note to the family: mood through the day, most-repeated questions, missed medicines, anything worrying.
5	Medicine and meal reminders	Gentle, routine-based reminders set up by the family.

Additional features

#	Feature	What it does
6	Memory Lane	Daily session: shows an old family photo and asks one gentle question. The patient's answers are saved into a life-story book the family keeps.
7	Evening calm mode	Detects late-day agitation (sundowning). Switches to slower, softer speech, the patient's favorite old songs, a familiar photo or family voice message, and alerts the family.
8	"I'm lost" safe return	On phrases like "Where am I?" or "I want to go home", shows home address and house photo, calls the first contact, and shares location (with family consent).
9	Family voice messages	Real recordings from family, tagged for the right moments (morning, meals, evening calm). No voice cloning.
10	Weekly doctor report	Trends over the week — repetition, night confusion, mood — in plain words with a simple chart. Clearly labeled as observations, not a diagnosis.
11	Visitor briefing	Before a visit, a relative gets today's mood, recent topics, and tips on what to talk about and what to avoid.
12	Scam and stranger shield	Patient taps "Someone is asking me something". The companion listens to that moment and warns if it hears requests for money, bank details, OTPs, or secrecy — then alerts family.

4. Architecture

Three parts

- **Family setup portal (web):** people (name, relation, photos, memories), daily routine, life story notes, favorite songs, voice messages, home address and house photo, emergency contacts, guidance for sensitive questions, consent settings.

- **Patient app (tablet, voice-first):** large buttons and text, calm design. Main actions: Talk, Show a face, Today's routine, plus the "Someone is asking me something" button.
- **Backend:** face matching, memory store, event log, alert system, and Nemotron agents running on Nebius.

Five shared building blocks

Most features are small add-ons once these five pieces exist. Build them all in week 1.

Building block	Used by
Family portal data	Every feature
Event log (every conversation, question, mood signal, reminder outcome — timestamped)	Daily digest, calm mode, doctor report, visitor briefing, scam shield
Alert system (one function that messages family; email or Telegram bot is enough for the demo)	Safety words, "I'm lost", calm mode, scam shield
Media player (photos, songs, voice messages)	Memory Lane, calm mode, family voice messages
Nemotron agents	Live conversation and all written reports

Model plan

Job	Model / tool	Notes
Live conversation	NVIDIA Nemotron 3.5 Lightning (Nebius Token Factory)	Fast, low-cost model suited to always-on agents.
Digest, doctor report, visitor briefing	NVIDIA Nemotron 3 Super (Nebius Token Factory)	Stronger reasoning; runs once per day/week over the event log.
Face matching	Open-source face-embedding library on our own server	More accurate than an LLM for identity, and face data stays private. Low confidence = no guess.
Scene understanding (optional)	NVIDIA vision model on Nebius AI Cloud, or a vision model on Token Factory	Check availability first; NVIDIA text models on Token Factory may be text-only.
Voice	Browser speech-to-text / text-to-speech for the demo	Upgrade to an NVIDIA speech model if time allows.

5. Safety and ethics principles

- Short, calm sentences; one idea at a time. Never argue with or harshly correct the patient.
- Never pretend to be a family member. No voice cloning — real recordings only.
- Never give medical advice or a diagnosis. The doctor report is observations only.
- Words like "pain", "fell", "lost", or "scared" trigger an immediate family alert.
- Face recognition never guesses: "I'm not sure who this is — shall I ask your daughter?"
- Location sharing and the scam shield only work with family consent; the shield listens only when the patient taps it.
- Family-installed and family-controlled. Face data and memories stay on our own server, not shared.

6. Who does what

Person 1 — Frontend and design	Person 2 — AI and backend
• Family portal (all data fields) • Patient app: camera, talk button, routine view, calm-mode screen, "I'm lost" screen, scam-shield button • Family views: digest, weekly report, visitor briefing, life-story book • Demo video visuals	• Nebius setup and Nemotron agents • Face matching • Memory store and event log • Alert system • Detection logic: safety words, calm mode, "I'm lost", scam patterns • All prompts, especially the dementia-friendly conversation rules

7. Week-by-week schedule

Week	Focus	Tasks
1 Oct 3–9	Building blocks	Nemotron 3.5 Lightning and Super working on Nebius • face matching from 3 photos • full family portal • event log and alert system • talk to one real carer or doctor and record a quote (with permission)
2 Oct 10–16	Patient experience	Recognition with context • repeated-question answers with memory lookup • reminders • painful-question guidance • family voice messages • "I'm lost" safe return
3 Oct 17–23	Intelligence layer	Daily digest • evening calm mode • Memory Lane and life-story saving • visitor briefing • weekly doctor report
4 Oct 24–30	Shield, polish, submit	Scam shield (Oct 24–25) • full test run with someone acting as the patient • design polish • demo video by Oct 27 • Devpost page, feedback, README • **submit by Oct 29**

Cut order if you fall behind: scam shield, then weekly doctor report, then visitor briefing. A polished core with eight or nine features beats a shaky twelve.

8. Demo video (about 3 minutes)

Time	Scene
0:00–0:20	The problem: a real carer's quote, plus the scale of dementia worldwide.
0:20–0:50	Family setup: adding Grandma's grandson with photos and a memory.
0:50–1:25	Recognition: Grandma points the tablet at Rohan; the companion answers warmly with context.
1:25–2:00	The painful question: "Where is my husband?" — the family's chosen gentle response. Emotional peak.
2:00–2:20	Evening calm mode: soft voice, an old favorite song, family alerted.
2:20–2:35	Scam shield catches a request for an OTP.
2:35–2:50	Montage: Memory Lane, "I'm lost", voice messages, digest, doctor report, visitor briefing.
2:50–3:00	Under the hood: Nemotron on Nebius, private face data, family owns everything.

9. Submission checklist

- Runs on Nebius Token Factory or Nebius AI Cloud
- Uses at least one NVIDIA open model — named clearly in the writeup
- Working demo video
- Public GitHub repo with setup instructions
- Feedback on the Nebius and NVIDIA tools used
- Entered in the Personal AI track
- Submitted by Oct 29 (hard deadline Oct 30, 10:00am PDT / 10:30pm IST)

10. After the hackathon

- Fully offline version on a small NVIDIA Jetson device for even stronger privacy (Personal AI track winners receive a Jetson Orin Nano).
- Paid caregiver shift notes and multi-carer handoff.
- Pilots with dementia care homes and support organizations.